

Math 4400, Fall 2016 Quiz 6 Name: _____

Calculators, cell phones, or notes are not allowed. The problems are worth 5 pts.

- (1) Solve $x^{27} = 7$ modulo 55 for some $0 \leq x \leq 54$.

$\varphi(55) = \varphi(5)\varphi(11) = 40$. Since $1 = 27 \cdot 3 - 2 \cdot 40$, it follows that $x = 7^3 = 343 \cong_{55} 13$.

- (2) Prove that if $n = p^k$ where p is prime, then $\sum_{d|n} \varphi(d) = n$.

Since the divisors of p^k are p^i for $0 \leq i \leq k$ and since $\varphi(1) = 1$ and $\varphi(p^i) = p^i - p^{i-1}$ for $i \geq 1$, we have

$$\sum_{d|n} \varphi(d) = \sum_{i=0}^k \varphi(p^i) = 1 + \sum_{i=1}^k (p^i - p^{i-1}) = \sum_{i=0}^k p^i - \sum_{i=1}^k p^{i-1} = \sum_{i=0}^k p^i - \sum_{j=0}^{k-1} p^j = p^n.$$

Without the fancy $\sum$ this reads as

$$\begin{aligned} \sum_{i=0}^k \varphi(p^i) &= 1 + (p-1) + (p^2-p) + (p^3-p^2) + \dots + (p^{k-1}-p^{k-2}) + (p^k-p^{k-1}) = \\ &= 1 + p + p^2 + p^3 + \dots + p^{k-1} + p^k - (1 + p + p^2 + \dots + p^{k-1}) = p^k. \end{aligned}$$